

6. Suppose we collect data for a group of students in a statistics class with variables X_1 = hours studied, X_2 = undergrad GPA, and Y = receive an A. We fit a logistic regression and produce estimated coefficient, $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$.

- (a) Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.
- (b) How many hours would the student in part (a) need to study to have a 50 % chance of getting an A in the class?

$$(a) \text{ logistic regression } \Rightarrow \text{logit}(p) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \\ = -6 + (0.05 \times 40) + (1 \times 3.5) = -0.5$$

$$p = \sigma(-0.5) = \frac{1}{1 + e^{0.5}} = \frac{1}{1 + 1.6} \approx 0.3775 \Rightarrow 37.8\% \quad \#$$

$$(b) \text{ 50\% chance } \Rightarrow p = 0.5 = \frac{1}{1 + e^{-z}} \Rightarrow z = 0$$

$$0 = -6 + (0.05 \times x_1) + (1 \times 3.5)$$

$$-0.05 x_1 = -2.5$$

$$x_1 = \frac{2.5}{0.05} = 50 \Rightarrow \text{need study 50 hours.} \quad \#$$